

Physics of Semiconductor Devices

ECE210: PSD



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

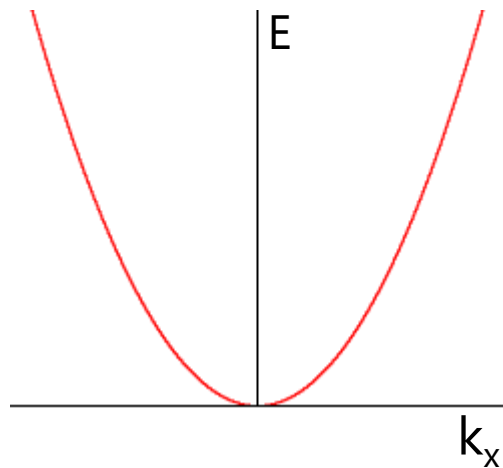
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Week 06



Free electron theory of metals

**Simple approximation:
treat electrons as free to
move within the crystal**



Free electron properties

Free electron Hamiltonian has
only kinetic energy operator:

$$E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

Free electrons are plane waves

$$\psi = A e^{\pm i k x}$$

with:

Momentum:

$$i\hbar \frac{\partial \psi}{\partial x} = \pm \hbar k \psi$$

Energy:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = \frac{\hbar^2 k^2}{2m} \psi$$

Group velocity:

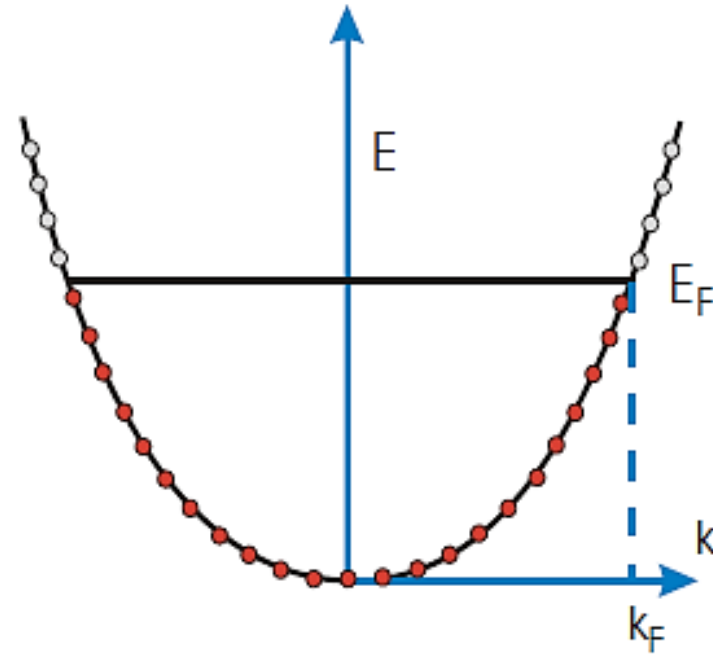
$$\frac{\partial \omega}{\partial k} = \frac{1}{\hbar} \frac{\partial E}{\partial k} = \frac{\hbar k}{m}$$

k-space picture and the Fermi Surface

T=0 states filled up to E_F

$$\therefore \frac{\hbar^2 k^2}{2m} = E_F$$

$$\therefore k_F = \sqrt{\frac{2mE_F}{\hbar^2}}$$



Merits and demerits of quantum free electron theory



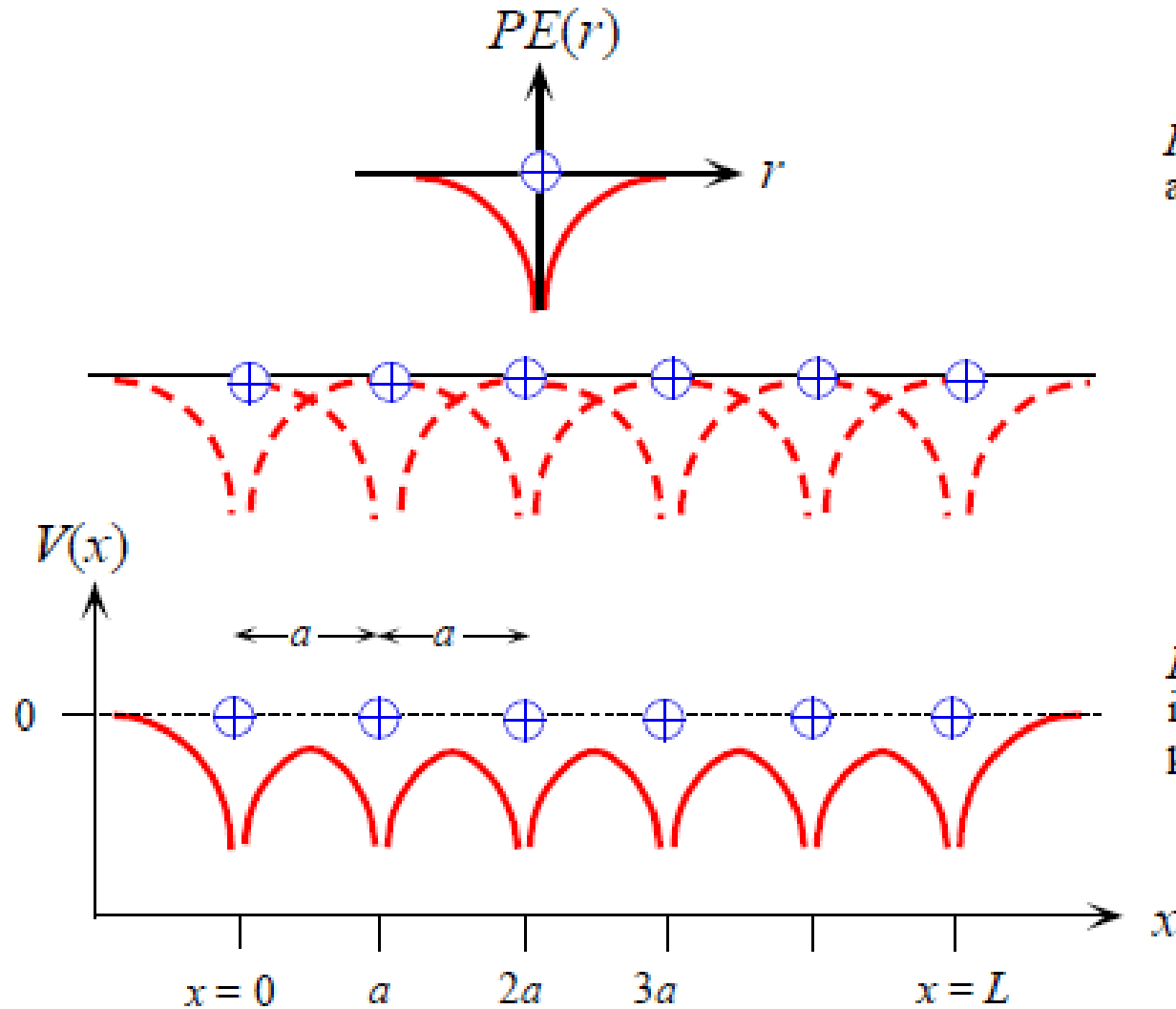
Merits of quantum free electron theory

1. It successfully explains the electrical and thermal conductivity of metals.
2. It can explain the Thermionic phenomenon.
3. It explains temperature dependence of conductivity of metals.
4. It can explain the specific heat of metals.
5. It explains magnetic susceptibility of metals.
6. It can explain photo electric effect, Compton Effect and black body radiation etc.
7. It gives the correct mathematical expression for the thermal conductivity of metals.

Demerits of quantum free electron theory

1. It is unable to explain the metallic properties exhibited by only certain crystals.
2. It is unable to explain why the atomic arrays in metallic crystals should prefer certain structures only.
3. This theory fails to distinguish between metal, semiconductor and Insulator.
4. It also fails to explain the positive value of Hall Co-efficient.
5. According to this theory, only two electrons are present in the Fermi level and they are responsible for conduction which is not true.

PE of electron in a crystal



PE of the electron around an isolated atom

When N atoms are arranged to form the crystal then there is an overlap of individual electron *PE* functions.

PE of the electron, $V(x)$, inside the crystal is periodic with a period a .

Bloch's Waves



If a periodic potential with period “ a ” can be defined as:

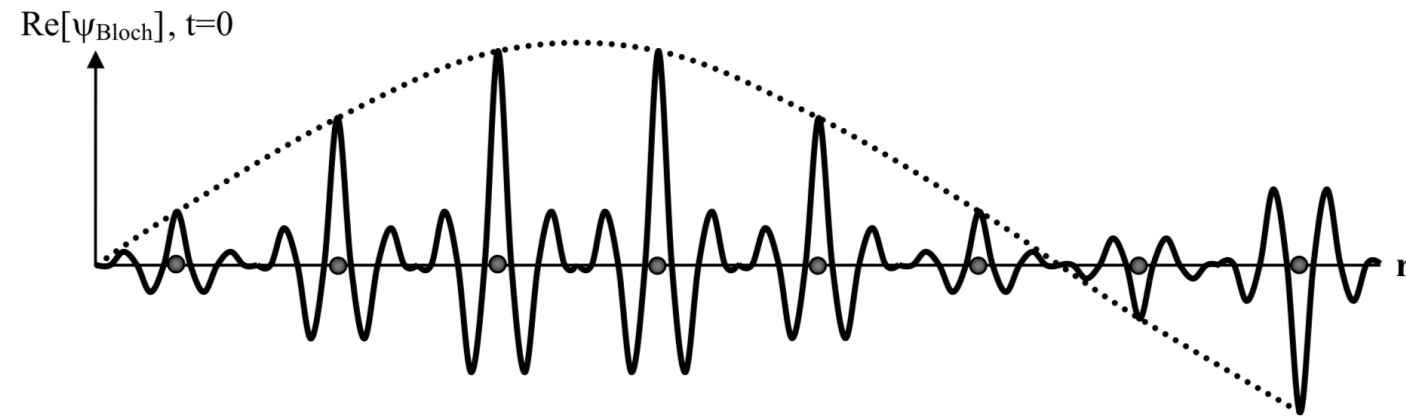
$$U(x + a) = U(x)$$

Then the wavefunction is periodic, and can be defined in terms of base function:

$$\Psi(x + a) = e^{ika} \Psi(x)$$

$$\Psi(x) = e^{ikx} u(x)$$

- a can be replaced by na



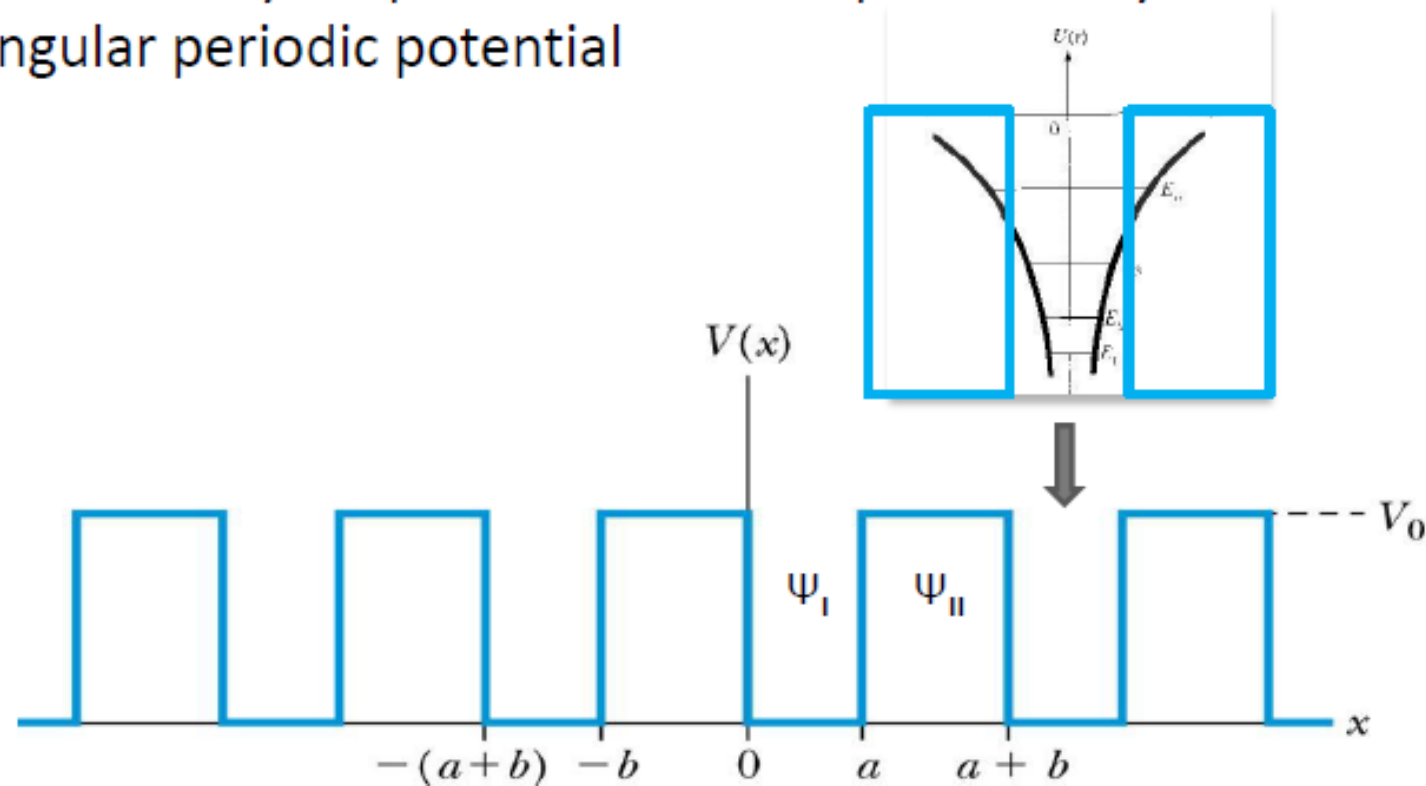
The application of Bloch's theorem helps explain the formation of valence band and conduction bands in a semi-conductor.

Kronig Penny model



Kronig-Penney Model

- Approximate crystal periodic Coulomb potential by rectangular periodic potential



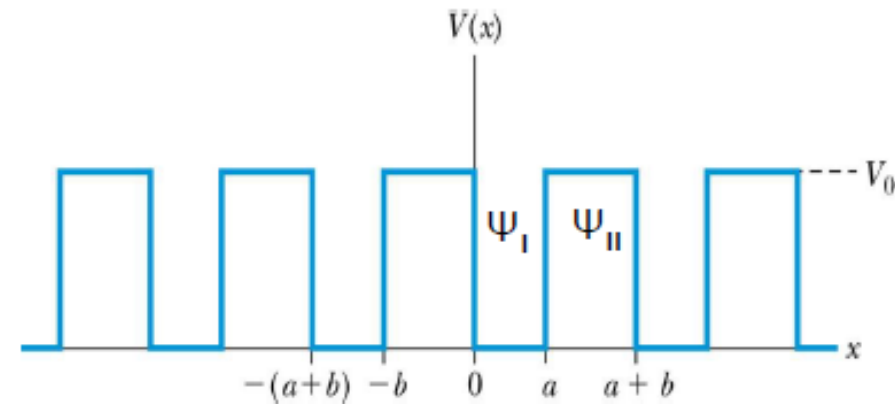
Kronig Penny model



Wavefunction Periodic Boundary Conditions

$$\Psi_I(0) = \Psi_{II}(0)$$

$$\frac{d\Psi_I(0)}{dx} = \frac{d\Psi_{II}(0)}{dx}$$



$$\Psi_I(a) = e^{ik(a+b)} \Psi_{II}(-b)$$

$$\frac{d\Psi_I(a)}{dx} = e^{ik(a+b)} \frac{d\Psi_{II}(-b)}{dx}$$

Reference: Neamen chapter 3 (page 63-70) for complete solution and its graphical representation

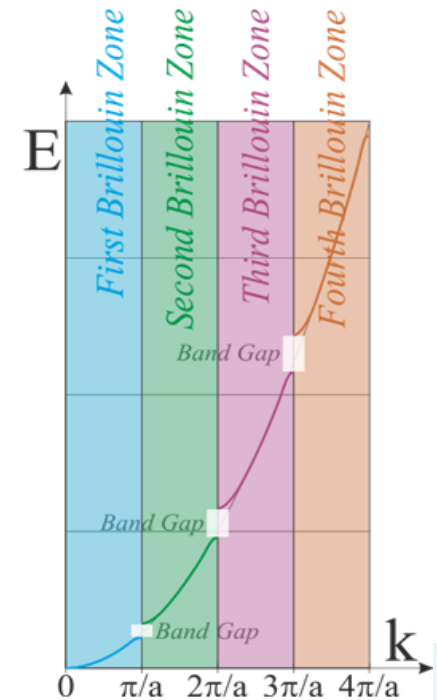
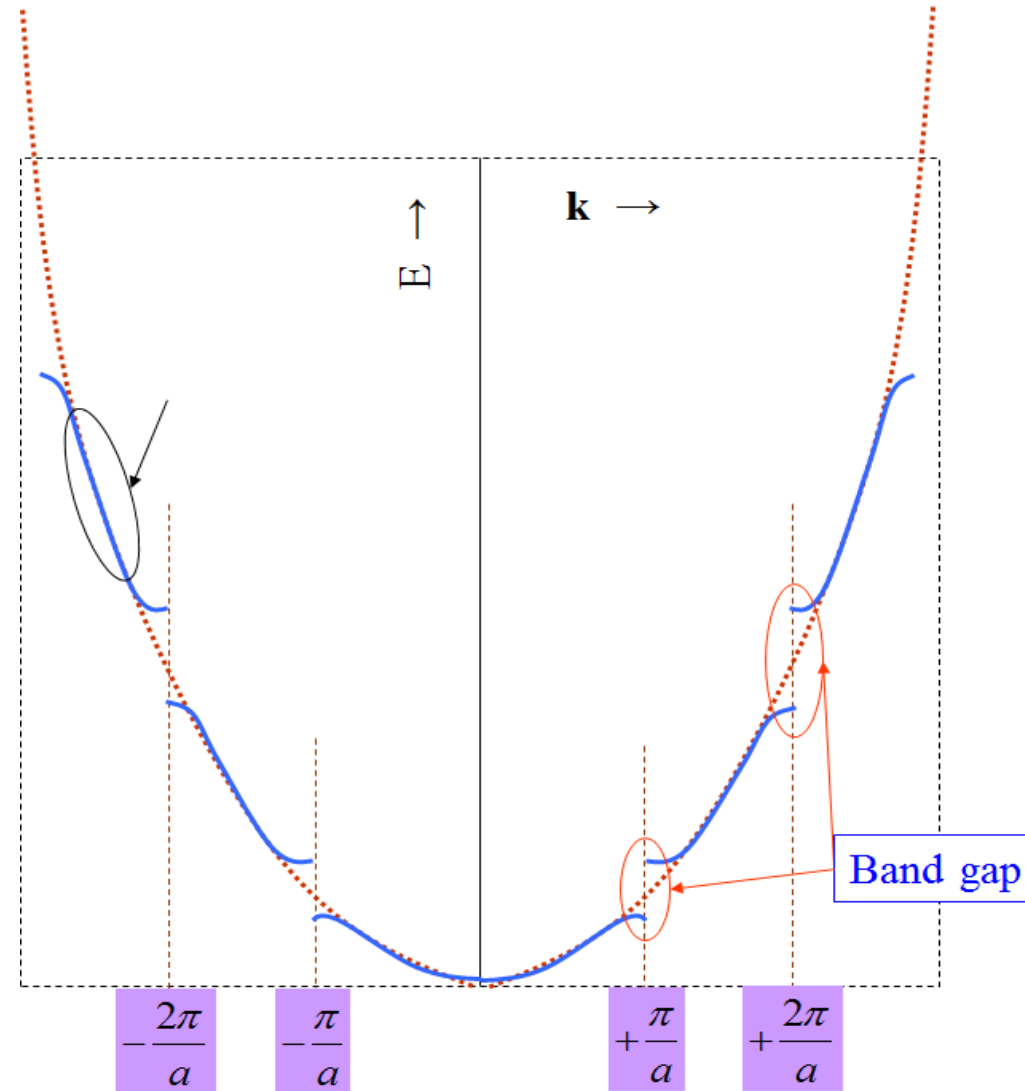
Energy dispersion in Kronig Penny model



By requiring the wavefunction and its derivative to be continuous everywhere, one finds energy levels that are grouped into **bands** separated by **energy gaps**. The gaps occur at

$$ka = \pm n\pi$$

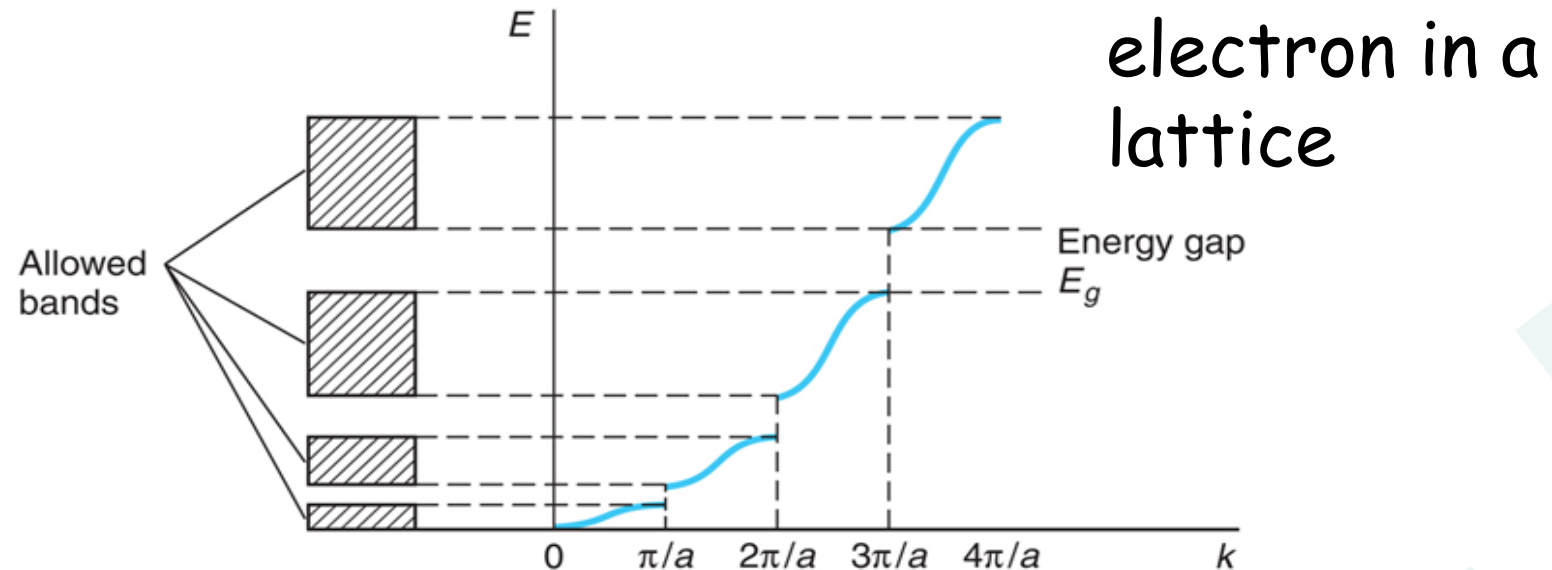
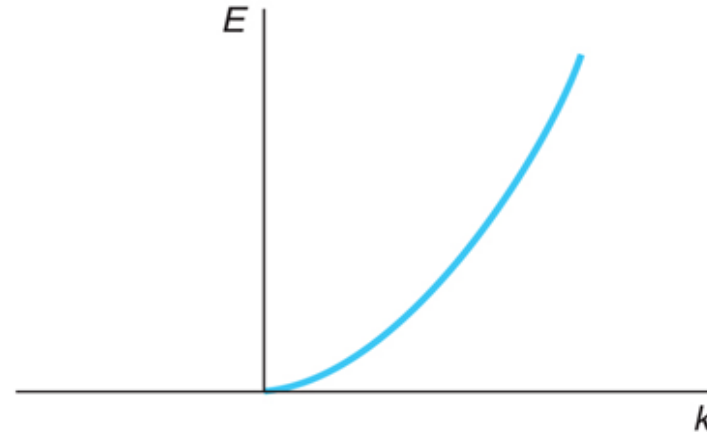
The energy gaps are basically energy levels that cannot occur in the solid



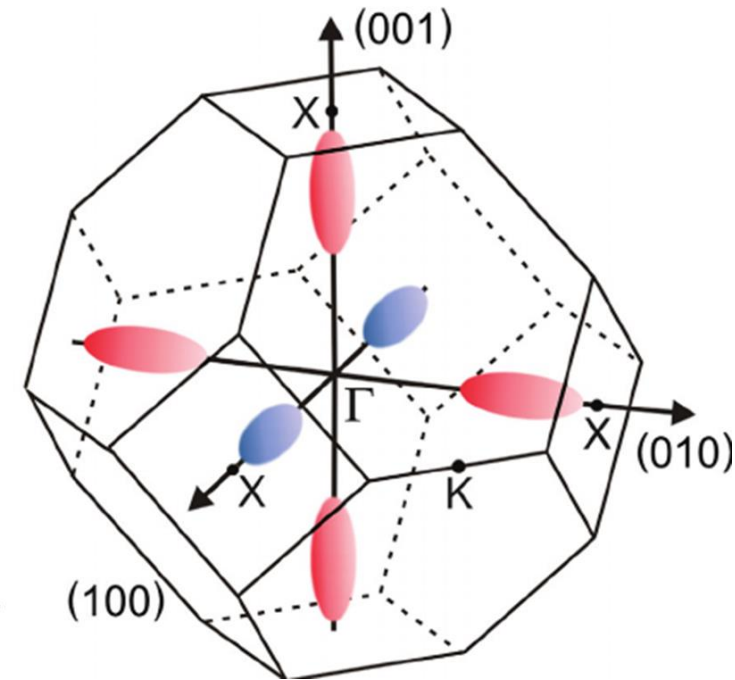
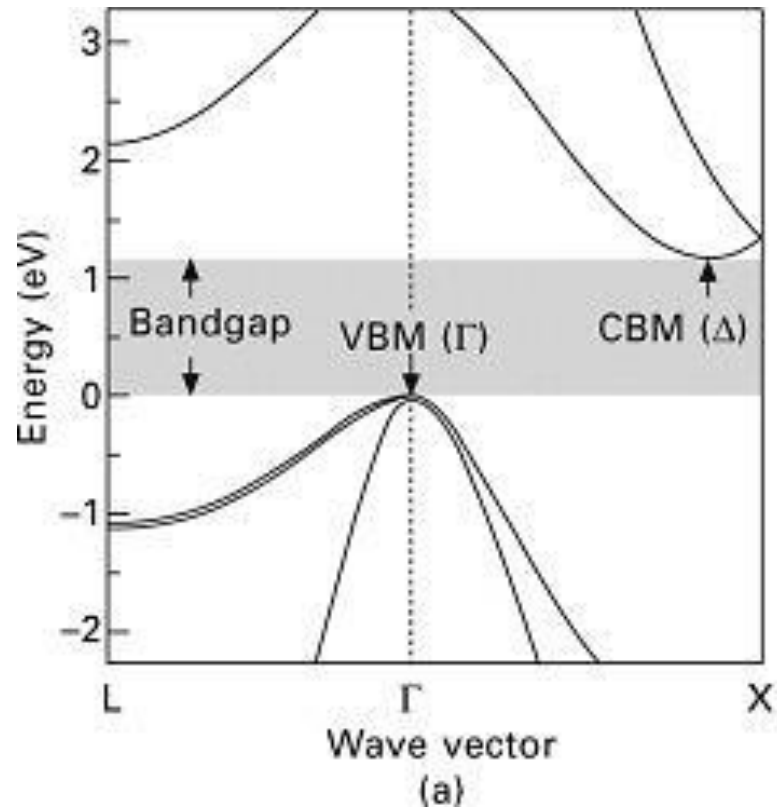
Electrons in a lattice



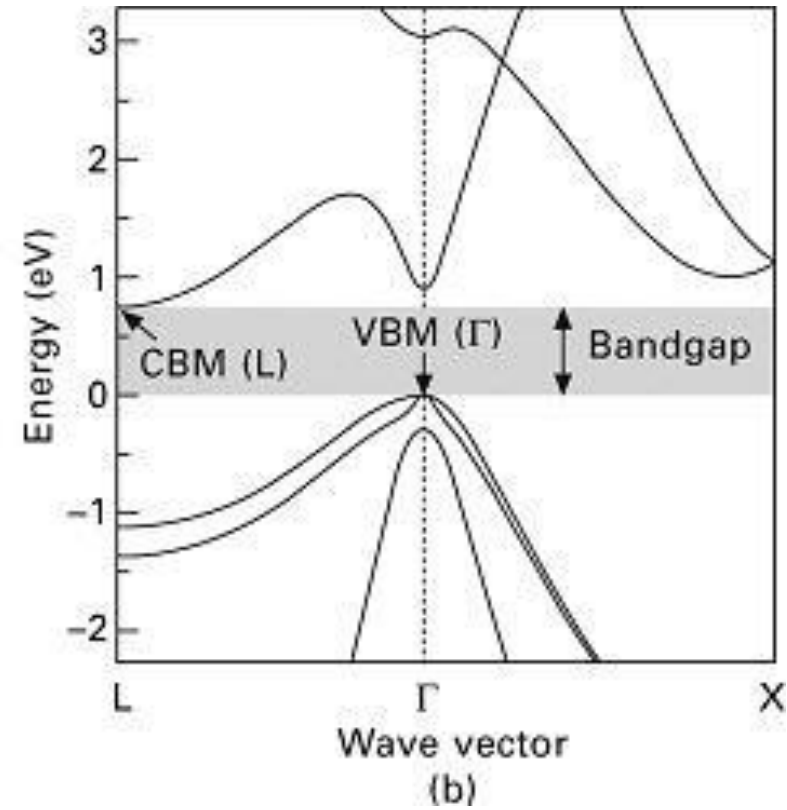
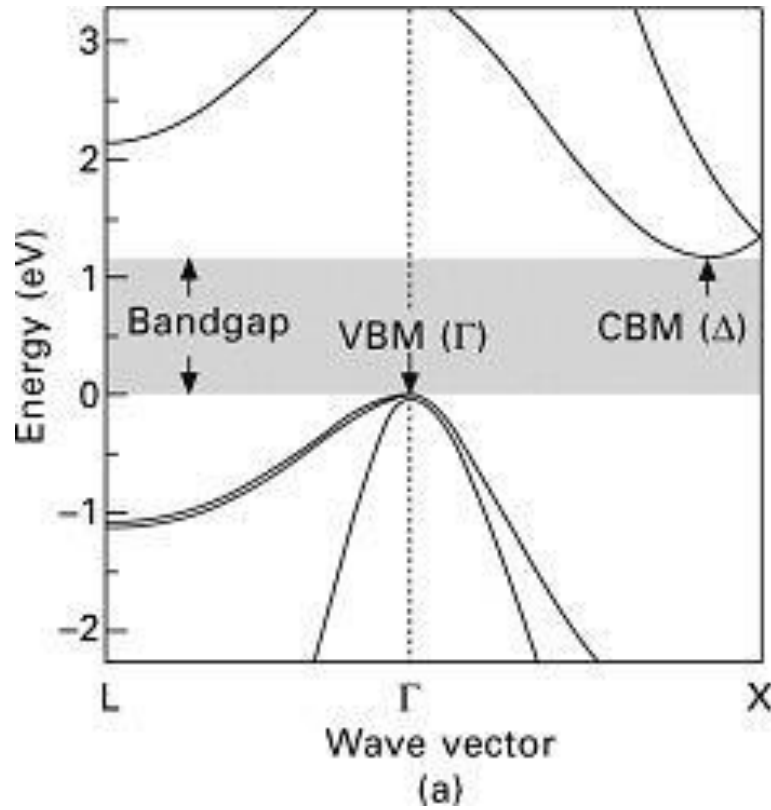
Completely free
electron



Band structure of silicon

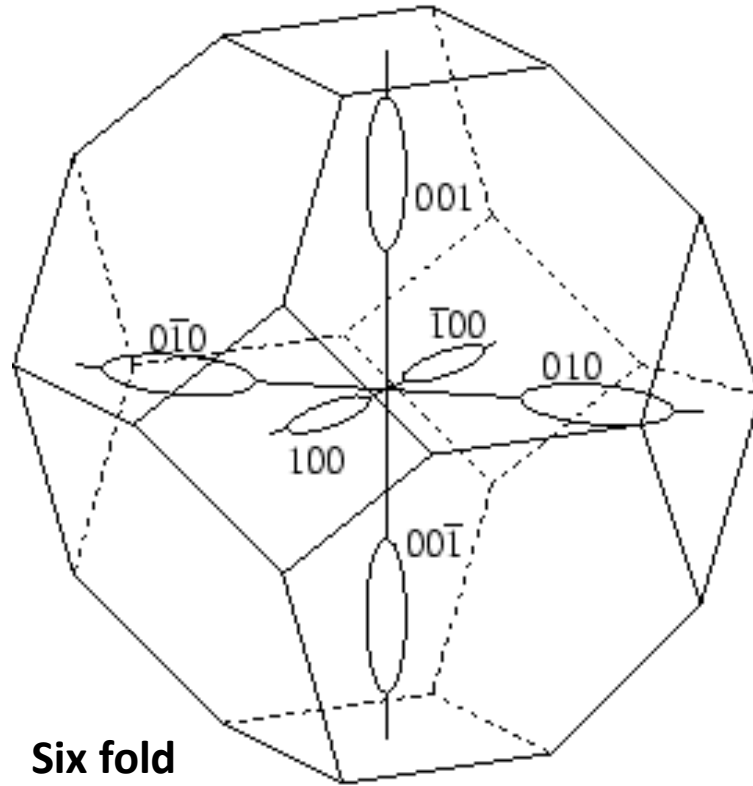


Band structure of silicon and germanium

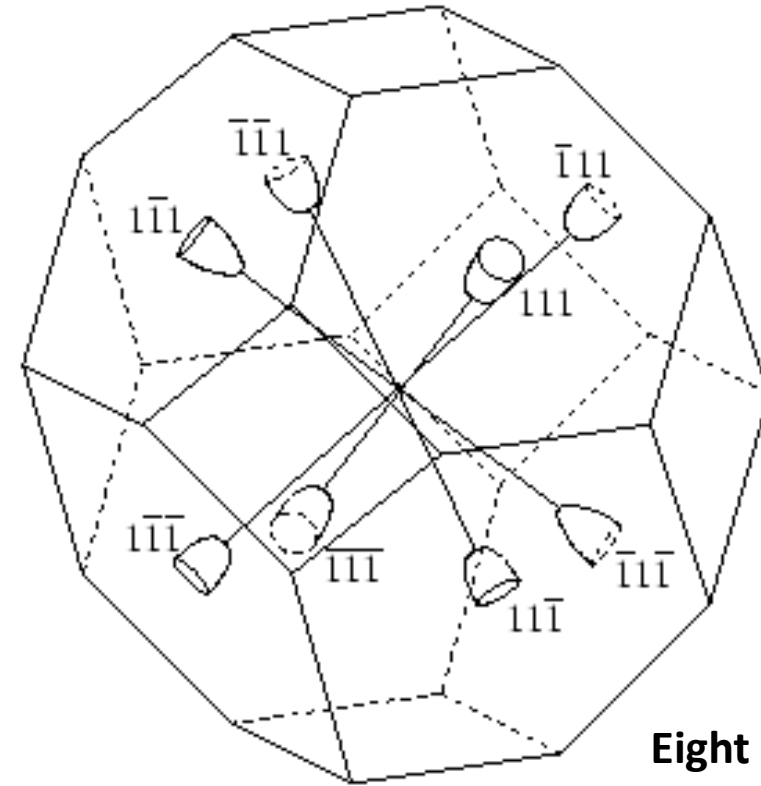


Energy valleys in silicon and germanium BS

The first Brillouin zone



X valleys of Si

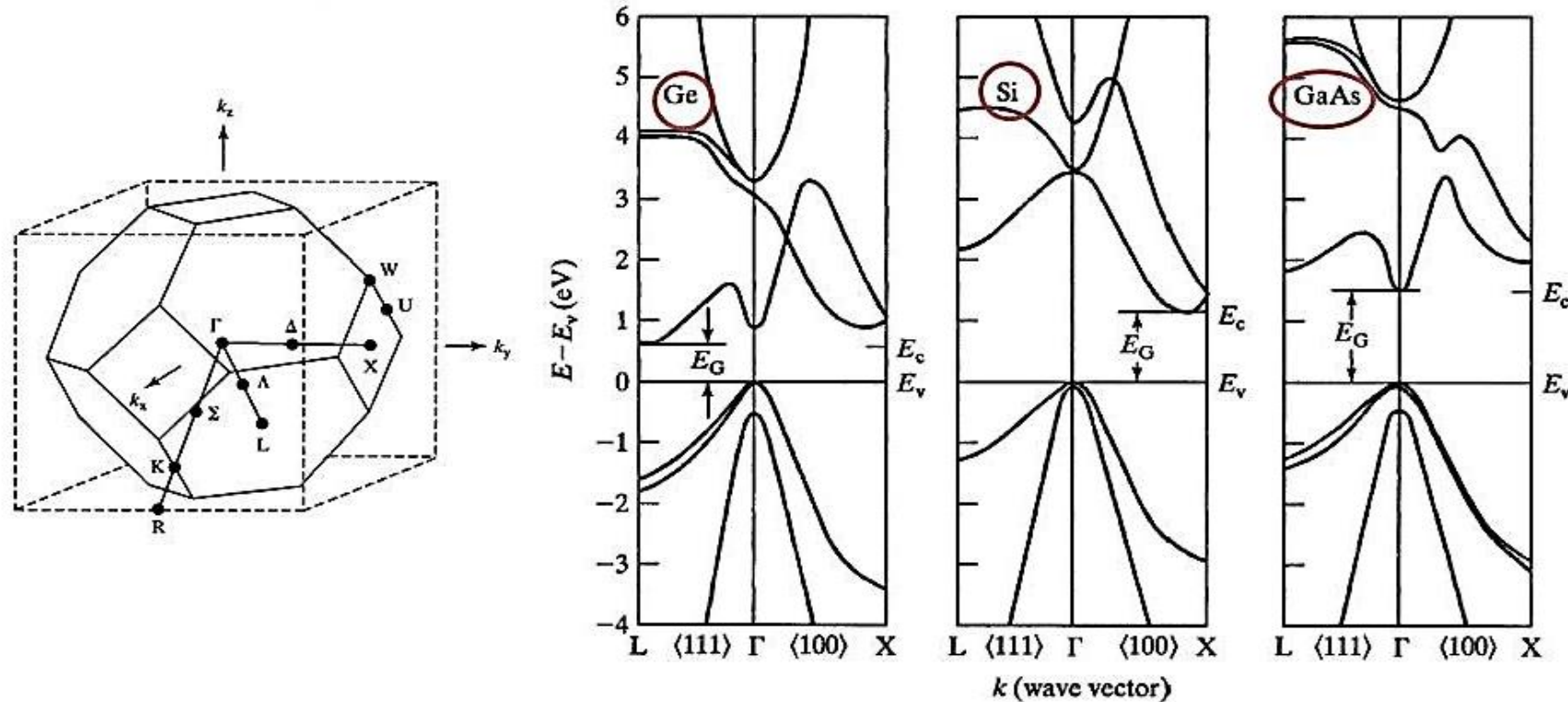


L valleys of Ge

Direct & Indirect semiconductor

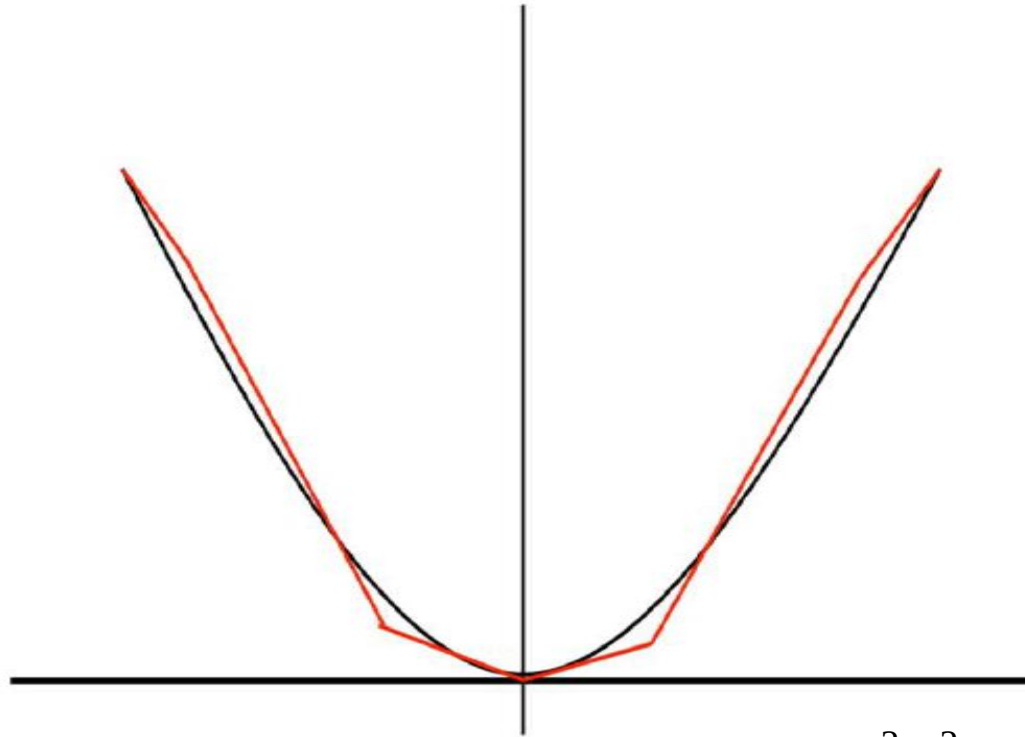


E-k Diagram



- 3 valence bands (light hole, heavy hole, split-off)
valence bands near $k=0$ is essentially $E \sim k^2$
- Minima may not be at zone center

Parabolic approximation



However, at the band edge we can approximate the E-k relationship to be parabolic

$$E = \frac{\hbar^2 k^2}{2m^*}$$

where $k^2 = k_x^2 + k_y^2 + k_z^2$

Effective mass



$$m^* = \frac{1}{\frac{1}{\hbar^2} \frac{d^2 E}{dk^2}}$$

